[인공지능]

Classification: Support Vector Machine

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Support-Vector Networks

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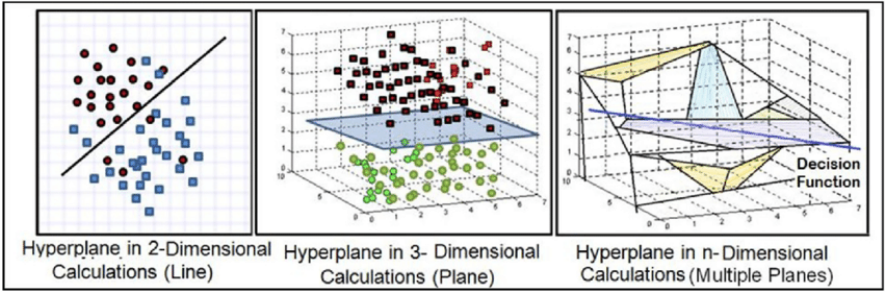
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Support Vector Machine

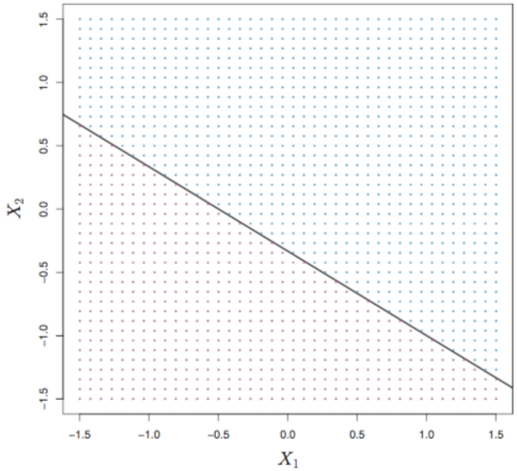
- **Support vector machine (SVM)** is a supervised learning model that constructs a decision boundary by maximizing the margin between classes, using support vectors, and extends to nonlinear boundaries through the kernel trick.
  - Linear
    - Hard Margin SVM (Maximal margin classifier)
    - Soft Margin SVM (Linear SVM)
  - Nonlinear
    - Kernel SVM

What is Hyperplane?

- **Hyperplanes** are decision boundaries that help classify data points.
- Hyperplane is an (p-1)-dimensional subspace of an p-dimensional space, which splits the space into two half-spaces.
  - In 2D, a hyperplane is a flat 1D subspace, a line.
  - In 3D, a hyperplane is a flat 2D subspace, a plane.


Decision
Function
Hyperplane in 2-Dimensional Calculations (Line)
Hyperplane in 3- Dimensional Calculations (Plane)
Hyperplane in n-Dimensional Calculations (Multiple Planes)

Separating Hyperplane (2D example)



- This hyperplane in 2 dimensions is the line:

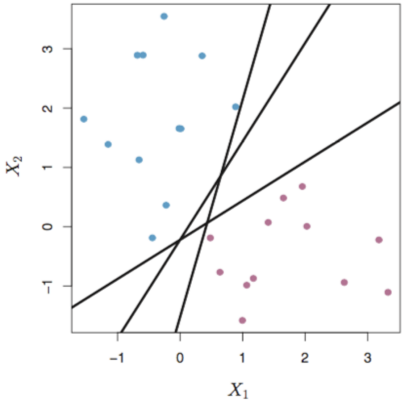
$$1 + 2X_1 + 3X_2 = 0$$

  - Blue region: $1 + 2X_1 + 3X_2 > 0$
  - Purple region: $1 + 2X_1 + 3X_2 < 0$

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Separating Hyperplane for Classification

- Idea: Use a separating hyperplane for binary classification.
- Key assumption: Classes can be separated by a linear decision boundary
- The hyperplane serves as a **decision boundary** that separates different classes.
- Notice that for a linearly separable dataset, there are many possible separating hyperplanes that divide the dataset into two classes (in fact, an infinite number).

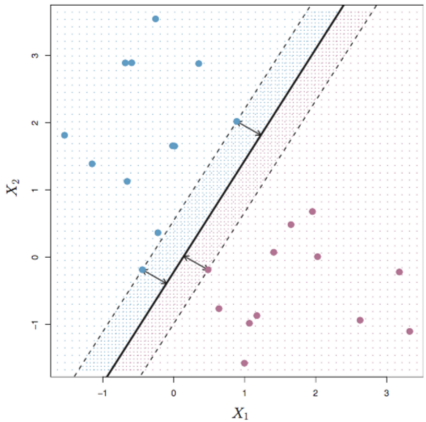


Maximal Margin Hyperplane

- Which of the infinite separating hyperplanes should we choose?

→ A natural choice is the **maximal margin hyperplane**, the separating hyperplane that is farthest from the training samples.

  - **Margin**: smallest distance between any training observation and the hyperplane
  - **Support vectors**: training observations whose distance to the hyperplane is equal to the margin



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Finding Maximal Margin Classifier (Hard Margin SVM)

- To find the maximal margin (M) hyperplane on data $(\vec{x}^{(i)}, y^{(i)})$, where $y^{(i)} \in \{-1, 1\}$, solve:

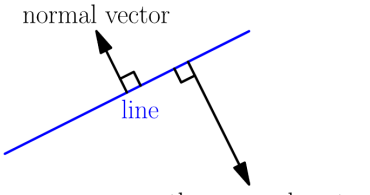
$$\max_{\beta_0,\ldots,\beta_p} M$$

$$\text{subject to } \sum_{j=1}^{p} \beta_j^2 = 1$$

$$y^{(i)} \left(\beta_0 + \beta_1 x_1^{(i)} + \cdots + \beta_p x_p^{(i)}\right) \geq M, \quad \forall i$$

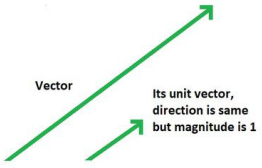
Background: Normal Vector & Unit Vector

- A vector that is perpendicular to a surface or a point on a surface.



Normal Vector

- A vector with a magnitude (or length) of 1.



Unit Vector

Why We Need Unit Vector?

- Both cases describe the same hyperplane

$$\text{Case 1: } \beta = (1,1),\ b = -5 \quad \Rightarrow \quad x_1 + x_2 - 5 = 0$$

$$\text{Case 2: } \beta = (2,2),\ b = -10 \quad \Rightarrow \quad 2x_1 + 2x_2 - 10 = 0$$

- Consider the training sample:

$$\mathbf{x}^{(1)} = (2,2), \quad y^{(1)} = +1.$$

- Without constraints of unit vector

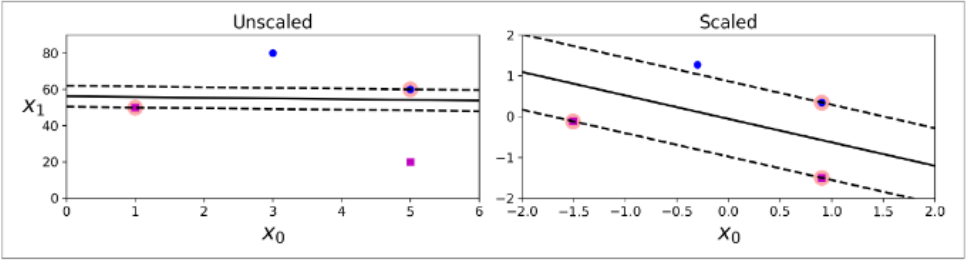
$$\text{Case 1: } M = y^{(1)}\left(\beta^\top \mathbf{x}^{(1)} + b\right) = (+1)\,(2+2-5) = -1,$$

$$\text{Case 2: } M = y^{(1)}\left(\beta^\top \mathbf{x}^{(1)} + b\right) = (+1)\,(4+4-10) = -2.$$

$\rightarrow$ While the hyperplane itself remains unchanged, the margin value varies, leading to an inconsistency.

Importance of Data Scale

- Unscaled data: Different feature ranges distort the normal vector, leading to an incorrectly defined margin.
- Scaled data: Features are normalized, making the norm meaningful and allowing the SVM to find the correct maximal margin hyperplane.



Feature Scaling Methods

- Scale each feature separately to reduce differences in measurement units.
- Standardization (Z-score)

$$\tilde{x}_j = \frac{x_j - \mu_j}{\sigma_j}$$

  - Centers the feature at zero and scales it to unit variance.
  - Does not restrict values to a fixed range.

- Min–Max Normalization

$$\tilde{x}_j = \frac{x_j - x_j^{\min}}{x_j^{\max} - x_j^{\min}}$$

  - Maps the training feature values to $[0, 1]$.
  - Preserves relative positions within the training range.

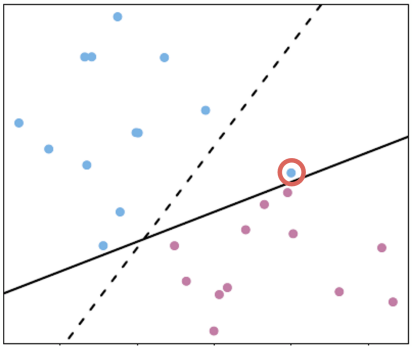
Where: x_j is the original value of feature j; μ_j and σ_j are its training mean and standard deviation; $x_j^{\min}$ and $x_j^{\max}$ are its training minimum and maximum.

- Compute scaling statistics from the training data only, and reuse them for validation and test data.
- tandardization is a common starting point for SVM.

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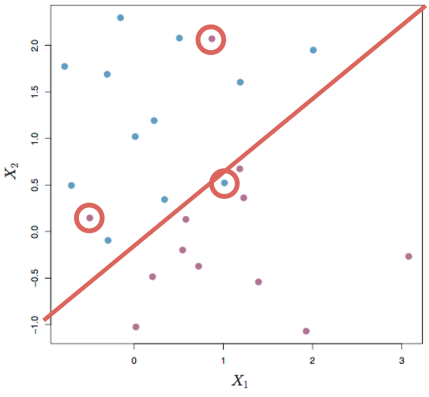
Disadvantage of the Maximal Margin Classifier

- Can be sensitive to individual observations
- May overfit training data



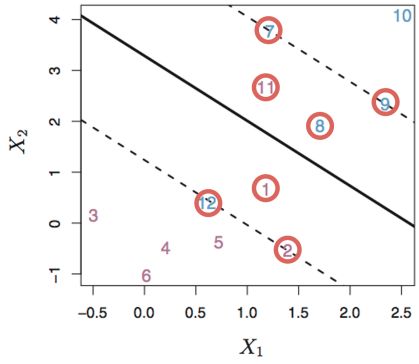
Soft Margin SVM

- Like the maximal margin classifier, it looks for a hyperplane to perform classification.
- However, training samples are allowed to be on the wrong side of the margin or hyperplane.
- This hyperplane almost separates the classes using a **soft margin**.



Support Vectors of Soft Margin SVM

- Points lying directly on the margin, or on the wrong side of the margin for their class.
- Some points are allowed to violate the margin (1, 8, 11, 12).



Finding Soft Margin SVM

- To find the soft margin SVM hyperplane, solve:

$$\max_{\beta_0,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n} M$$

$$\text{subject to} \quad \sum_{j=0}^{p} \beta_j^2 = 1$$

$$y^{(i)} \left(\beta_0 + \beta_1 x_1^{(i)} + \cdots + \beta_p x_p^{(i)} \right) \geq M(1 - \epsilon_i), \quad \forall i$$

$$\sum_{i=1}^{n} \epsilon_i \leq C, \quad \epsilon_i \geq 0, \quad \forall i$$

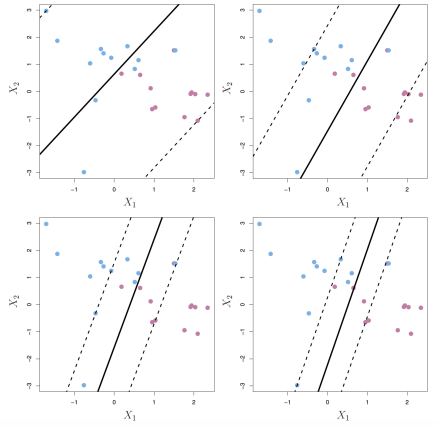
- Slack variables ϵ_i allow for violations of the margin.
  - $\epsilon_i = 0$: training point is on the correct side of the margin
  - $0 < \epsilon_i \leq 1$: training point violates the margin
  - $\epsilon_i > 1$: training point is misclassified (wrong side of hyperplane)

Penalty Parameter C

- **Penalty parameter** C controls the trade-off between margin width and the tolerance of classification errors.
- C is a hyperparameter that we tune using cross-validation.
- Note that it must be non-negative.
- If $C = 0$, we recover the maximal margin classifier (if one exists).
- As C goes from small to large, there is a bias-variance tradeoff.

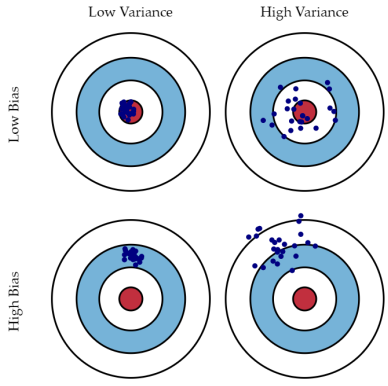
How C Influences the Margin

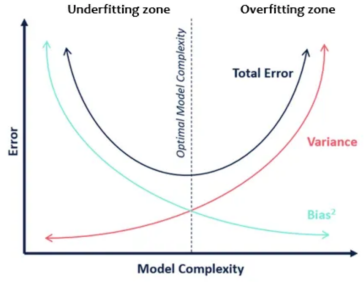
- Small C
  - Large margin
  - Allows for some errors, leading to a simpler model (Underfitting).
  - High bias, low variance
- Large C
  - Small margin
  - Tries to minimize errors and separates the data more tightly (Overfitting).
  - Low bias, high variance

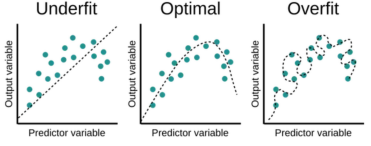


Bias-Variance Tradeoff

- Bias-Variance Tradeoff[1]







Soft Margin SVM and Hinge Loss

- Let

$$f_\beta(\mathbf{x}) \;=\; \beta_0 + \sum_{j=1}^{p} \beta_j x_j, \qquad \|\boldsymbol{\beta}\|_2^2 \;=\; \sum_{j=1}^{p} \beta_j^2.$$

- Fixing $M = 1$

$$y^{(i)}\, f_\beta\Big(\mathbf{x}^{(i)}\Big) \;\geq\; 1 - \epsilon_i, \qquad \epsilon_i \geq 0 \quad (\forall i),$$

- We obtain

$$\epsilon_i \;\geq\; \max\Big\{\, 0,\; 1 - y^{(i)} f_\beta\Big(\mathbf{x}^{(i)}\Big) \,\Big\}.$$

- Hinge loss interpretation

$$\ell_{\text{hinge}}\Big(y^{(i)}, f_\beta(\mathbf{x}^{(i)})\Big) = \max\Big\{0,\; 1 - y^{(i)} f_\beta(\mathbf{x}^{(i)})\Big\}.$$

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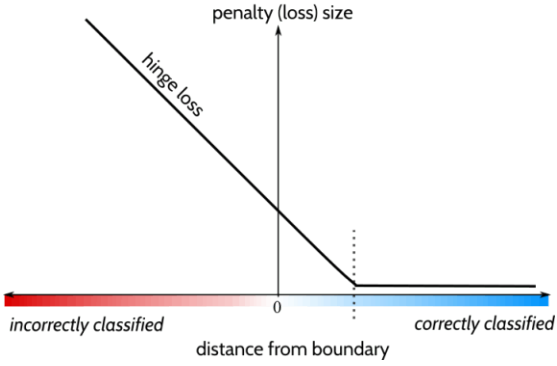
- Equivalent optimization problem (without explicit ϵ_i)

$$\min_{\beta_0,\ldots,\beta_p} \ \frac{1}{2}\,\|\boldsymbol{\beta}\|_2^2 \ + \ C\sum_{i=1}^{n} \max\Big\{0,\ 1 - y^{(i)} f_\beta\Big(\mathbf{x}^{(i)}\Big)\Big\}.$$

- Equivalently, with slack variables:

$$\min_{\beta_0,\ldots,\beta_p,\ \epsilon_1,\ldots,\epsilon_n} \ \frac{1}{2}\,\|\boldsymbol{\beta}\|_2^2 \ + \ C\sum_{i=1}^{n} \epsilon_i \quad \text{s.t.} \quad y^{(i)} f_\beta(\mathbf{x}^{(i)}) \geq 1 - \epsilon_i,\ \epsilon_i \geq 0.$$

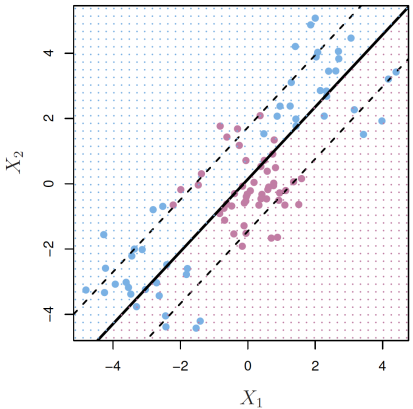
Hinge Loss



Source

Linear Boundary Can Fail

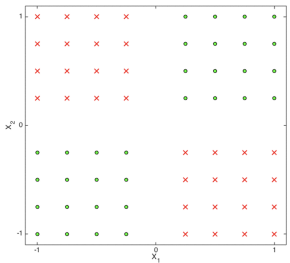
- Sometimes, a linear boundary simply won't work, no matter what value of C.



Expanding Feature Space

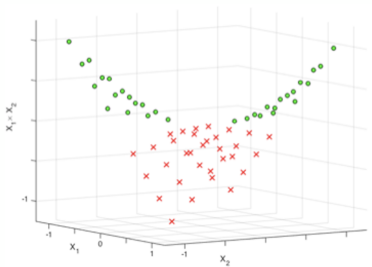
- Some datasets are not linearly separable, but they become linearly separable when transformed into a higher-dimensional space.

Original feature space



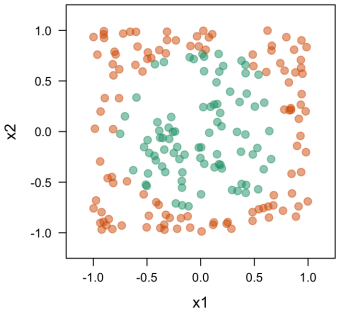
variables x_1, x_2

New feature space

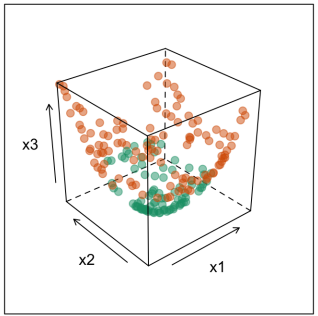


variables x_1, x_2, x_1x_2

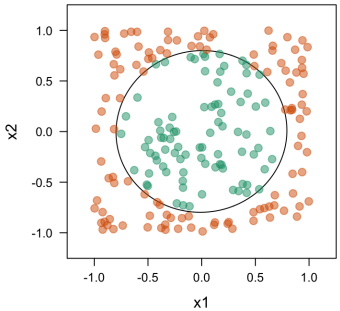
Original feature space



Enlarged feature space



Non-linear decision boundary



https://bradleyboehmke.github.io/HOML/svm.html

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- Suppose our original data has p features:

$$\vec{X} = (X_1, X_2, \ldots, X_p)$$

We can expand the feature space to include $2p$ features:

$$\vec{X} = (X_1, X_1^2, X_2, X_2^2, \ldots, X_p, X_p^2) = (\tilde{X}_1, \tilde{X}_2, \ldots, \tilde{X}_{2p})$$

- In the expanded space, the support vector classifier finds a linear hyperplane:

$$\beta_0 + \beta_1 \tilde{X}_1 + \beta_2 \tilde{X}_2 + \cdots + \beta_{2p} \tilde{X}_{2p} = 0$$

- In the original space, this becomes a nonlinear decision boundary:

$$\beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \cdots + \beta_{2p-1} X_p + \beta_{2p} X_p^2 = 0$$

Nonlinear SVM

- Extends the support vector classifier by using kernel functions to achieve non-linear decision boundaries.
- **Kernel function** K: generalization of inner product.
  - It takes in two arguments and implicitly computes their inner product in some feature space.
  - Kernels are an efficient computational approach to create non-linear decision boundaries.
  - They (implicitly) map data into a higher-dimensional space.
  - We then apply a support vector classifier in this high-dimensional space with a linear decision boundary (hyperplane).

Linear SVM and Nonlinear SVM

- It can be shown that a support vector classifier (linear) can be represented as:

$$f(\mathbf{x}) \;=\; \beta_0 + \sum_{i \in S} \alpha_i \, \langle \mathbf{x}^{(i)}, \mathbf{x} \rangle$$

- In nonlinear SVM, we replace the inner product with a kernel function:

$$f(\mathbf{x}) \;=\; \beta_0 + \sum_{i \in S} \alpha_i \, K(\mathbf{x}^{(i)}, \mathbf{x})$$

SVM Kernels

- **Linear kernel:**

$$K(\mathbf{x}, \mathbf{x}') = \langle \mathbf{x}, \mathbf{x}' \rangle$$

- **Polynomial kernel (degree p):**

$$K(\mathbf{x}, \mathbf{x}') = \big(1 + \langle \mathbf{x}, \mathbf{x}' \rangle\big)^p$$

- **Radial basis function (RBF) kernel:**

$$K(\mathbf{x}, \mathbf{x}') = \exp\Big(- \gamma \, \|\mathbf{x} - \mathbf{x}'\|^2 \Big)$$

SVM Summary

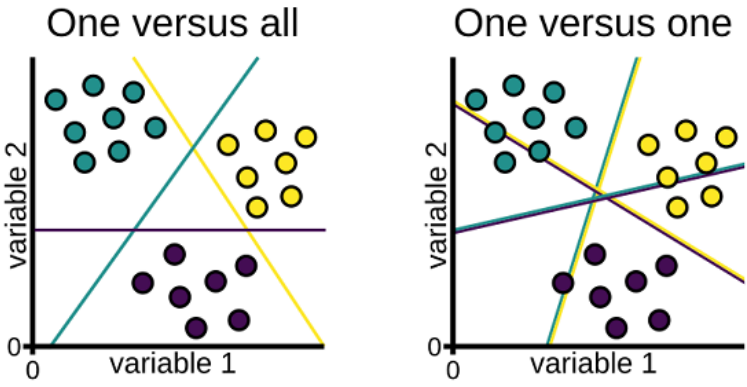
Pros:

- Regularization parameter C helps avoid overfitting
- Use of kernel gives flexibility in the shape of the decision boundary
- Optimization problem is convex — unique solution

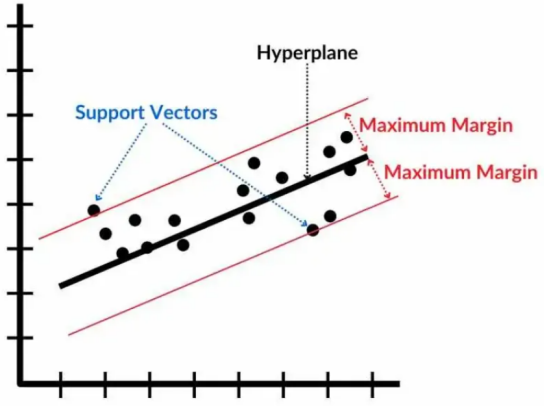
Cons:

- Must tune hyperparameters (e.g. C, kernel function)
- Must formulate as binary classification
- Difficult to interpret

Multi-class SVM



Support Vector Regression (SVR)



Thank You!

(Q & A)

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